

Title:

How does ICT development influence service exports in developing and emerging countries?

Abstract

This research paper considers the influence of information and communication technology (ICT) advancement on the export performance of services in 25 developing and emerging economies from 2009 to 2019 using panel data analysis. An aggregate ICT Index—mobile subscriptions, internet users, and broadband penetration—is fitted onto log service exports, GDP per capita, and population. Fixed and Random Effects models are estimated, of which the Hausman test favors the Fixed Effects model. Empirical evidence indicates that ICT growth highly increases the exports of services, and vice versa for GDP per capita and population. Policy suggestions emphasize the upgrading of ICT infrastructure, digital literacy, building robust governance regimes, and aligning ICT strategies with export and innovation policies to maximize the performance of services trade.

Introduction

As the global economy has been integrated, digital technology has emerged as the cornerstone of international production, trade, and economic transformation. The spread of information and communication technology has restructured the production and cross-border delivery of services from online banking to remote education and IT offshoring. As services have become increasingly digitized, it is crucial to understand how the evolution of ICT fosters export performance for attaining sustainable and inclusive growth.

According to the World Trade Organization (2024), over 54% of global service exports are delivered digitally and valued at around USD 4.2 trillion. Developing nations are catching up rapidly, with digital service exports growing at about 12% annually for the past decade.

Installation of ICT infrastructure—mobile networks, internet access, and broadband—has opened

up new opportunities for diversification of trade and integration in global value chains, facilitating ICT as a principal economic enabler.

Despite this development, the service exports-ICT relationship remains poorly researched in developing contexts. Advanced economies remain the central subject of examination at the expense of structural and institutional determinants of outcomes. The degree to which per capita GDP and population moderate the relationship is also uncertain.

To address these gaps, the following research questions are there which this paper aim at:

1. How does ICT development influence the growth of service exports in developing and emerging economies?
2. Why do some countries with rapid ICT expansion show weaker or negative ICT–service export relationships?
3. To what extent do GDP per capita and population shape this relationship?
4. Which countries exhibit the strongest and weakest ICT–export linkages, and what factors explain these variations?

By addressing these questions, the study offers an idea of how digital transformation can enhance global competitiveness and foster sustainable service trade growth in developing economies.

Literature Review

The fast-paced spread of information and communications technology has transformed world production systems, trade flows, and growth patterns, particularly in the services sector.

Increased digitalization of economies enabled services, heretofore regarded as non-tradable, to increasingly become exportable via online and technology-driven channels. ICT development, therefore, emerged as a crucial determiner of service sector competitiveness in developing and emerging economies. The subject is rooted in the general literature on technology-led growth and trade that focuses on how information infrastructure can help late-developers to access global markets and undergo structural change. ICT lowers transaction costs, allows for cross-border delivery of digital services, and raises the engagement of developing economies within global value chains, based on networking by the World Bank (2023) and WTO (2024).

The link between ICT and service exports can be explained through several economic frameworks. According to traditional trade theories, we get to know that technological progress shifts comparative advantage toward more information and skill-intensive sectors. The new trade theory and new economic geography highlight that ICT reduces fixed and variable trade costs, enabling more firms to access foreign markets. According to endogenous growth theory, it is considered a general-purpose technology, and ICT increases productivity through knowledge spillovers, innovation, and network externalities. To account for the decrease in communication barriers and information asymmetries, the gravity model of trade has also been leveraged to the service sector by the introduction of ICT-related variables. Overall, these theories suggest that ICT facilitates the service export ability through increased efficiency, connectivity, and access to customers around the world.

These theoretical claims mentioned above are supported by the strong evidence given with the help of empirical research. Li (2023), with their co-authors, have reaffirmed in their paper that digitalization greatly stimulates world service exports, especially through improved broadband and mobile internet connections. Wen et al. (2023), examining 131 countries' data between 2005 and 2020, depicted that digital service trade brings innovation and raises the tech intensity of trade. Also, Sultanova and Gnangnon (2024) presented evidence that ICT expansion helps to diversify exports in developing economies, with broadband quality playing the greatest role. The IMF (2023) and OECD (2024) also found additional evidence that investment in ICT has high multiplier impacts on service output and employment. Such studies as these, in fact, set ICT as the major cause of services trade competitiveness.

At the regional level, numerous studies have explored the ICT–service export relationship in specific contexts. Hanif and Zafar (2024) have taken South Asian economies into consideration and found that ICT infrastructure, human capital, and openness significantly complement service exports. Kumari et al. (2023) affirm long-run cointegration among ICT, financial development, and trade openness for developing countries, substantiating economic as well as digital powers' complementary role. Latin America, Kerner (2022) illustrated, was better positioned to endure shocks such as the COVID-19 pandemic since it boasted ICT-intensive service exporters that were digitally literate. Hossain and Lee (2022) highlight in their study that service exporting in African economies is significantly influenced by Internet penetration and bandwidth, albeit in a

moderating capacity related to institutional vulnerability. These findings show that ICT enhances service export but that its impact is complementary, capability-sensitive.

The overwhelming majority of empirical studies on ICT and service exports have used panel data regression techniques, specifically fixed effects (FE) and random effects (RE) models, to account for country and time variability. The Hausman test is used for deciding the appropriate model. Instrumental variable (IV) or system GMM is applied by some authors. One of them was Wen et al. (2023) to address potential endogeneity between ICT and exports, while others checked for long-run relationships employing panel cointegration models. Researchers have the tendency to construct an ICT index—either through normalization or PCA—by combining indicators such as mobile subscription, internet users, and broadband penetration. This provides a general picture of ICT development and evades the multicollinearity issue.

In most cases, these studies find that development in ICT benefits service exports, but the magnitude of the impact depends on aspects like infrastructure, human resources, and institutions. In developing countries, there are a lot of shortcomings, such as causation and heterogeneity.

Methodology

To investigate the influence of ICT development on the service sector in developing and emerging economies, this study uses a panel data model taking 25 countries over the period 2009–2019.

The econometric model, which is used in this paper, is as follows:

$$\text{Service Export}_{it} = \beta_0 + \beta_1 \text{ICT Index}_{it} + \beta_2 \text{GDP}_{it} + \beta_3 \text{Population}_{it} + \mu_i + \epsilon_{it}$$

where:

- i represents the country ($i=1, 2, \dots, 25$)
- t represents the time period ($t=2009, \dots, 2019$)
- μ_i captures unobserved country-specific effects, and
- ϵ_{it} is the error term

The 25 countries included are mentioned as: India, Indonesia, Kazakhstan, Argentina, Bangladesh, Kenya, Malaysia, Mexico, Morocco, Brazil, Chile, China, Colombia, Egypt (Arab Rep.), Ethiopia, Ghana, Pakistan, Peru, Philippines, Saudi Arabia, South Africa, Sri Lanka, Thailand, United Arab Emirates, Vietnam.

The study's variables—ICT Index, GDP per capita, and population—are both theoretically and empirically appropriate for ascertaining how service exports are promoted, especially in developing and emerging markets. The ICT Index is an indicator of technological capability and digital readiness, which has a fundamental bearing on productivity, reducing trade costs, and enabling service provision across borders. GDP per capita controls for the levels of economic development and infrastructure, while population accounts for demographic scale and possible outcomes for labor. The years that caught this study's period (2009-2019) represent a decade of convulsive digital transformation from the global financial crisis and are truncated in order to limit the distorting impact of the pandemic in terms of the data used. The period also assists with the reliability of the data because the extant datasets from the World Bank and WTO have provided grounds for reliability due to consistency in data collected from country to country.

Variable Description:

Variable	Type	Description	Expected Sign
Service Export	Dependent Variable	It reflects the performance and international competitiveness of the service sector.	
ICT Index	Independent Variable	Combination of three indicators: (1) mobile cellular subscriptions (per 100 people), (2) percentage of internet users, and (3) fixed broadband subscriptions (per 100 people). The index is constructed using a normalized average of these indicators.	Positive (+) — greater ICT development facilitates trade in digital and knowledge-based services.

GDP per capita	Control Variable	Measures economic development and productive capacity.	Positive (+) — higher income supports infrastructure and human capital necessary for exporting services.
Population	Control Variable	Captures market size and labor availability	Ambiguous (+/-) — may boost labor supply but could strain infrastructure if unmanaged.

Note: Service exports, GDP, and Population are converted to logarithms to match the ICT index

The hypothesis of this study is mentioned below:

- H_0 (Null Hypothesis): ICT development has no significant impact on service exports in developing and emerging countries. This is shown as: $H_0: \beta_1=0$
- H_1 (Alternative Hypothesis): ICT development positively influences service exports in developing and emerging countries. This is shown as : $H_1: \beta_1>0$

Given the dataset's cross-sectional (25 countries) and time-series (11 years) dimensions, a panel data regression approach is employed to capture both temporal and country-specific effects.

Two models are estimated:

1. Fixed Effects Model (FEM): Accounts for time-invariant, country-level heterogeneity that may be correlated with the explanatory variables.

$$\text{Service Export}_{it} = \alpha_i + \beta_1 \text{ICTIndex}_{it} + \beta_2 \text{GDP}_{it} + \beta_3 \text{Population}_{it} + \epsilon_{it}$$

2. Random Effects Model (REM): Treats unobserved country-fixed effects as random and orthogonal to the regressors.

$$\text{Service Export}_{it} = \alpha_i + \beta_1 \text{ICTIndex}_{it} + \beta_2 \text{GDP}_{it} + \beta_3 \text{Population}_{it} + \mu_i + \epsilon_{it}$$

In order to conclude which model is more suitable, the Hausman test is conducted. It said that: the Null Hypothesis (H_0) means that the Random effects model is appropriate, and, on the other hand, the Alternative Hypothesis (H_1) means the Fixed effects model is appropriate. Depending on the value of the Hausman test, we decide which model is the best. If the p-value is below 0.05, we reject the null in favor of the Fixed Effects Model, and if the p-value is above 0.05, we favor the Random Effects Model.

Prior to regression analysis, descriptive statistics (mean, median, standard deviation, minimum, maximum) summarize the data and indicate variable dispersion across countries and over time.

The correlation matrix shows potential for multicollinearity between explanatory variables, when applicable, particularly ICT components or GDP per capita and ICT development. If pairwise correlation coefficients are below 0.8, multicollinearity is not considered problematic.

It is assumed that the ICT Index will be positively and statistically significant with respect to service exports, suggesting that digitalization and communications improvements enhance developing economies' ability to export services. Similarly, GDP per capita is anticipated to be positive, indicative of greater economic capacity, while the population's effect could be complicated depending on the country's structure or labor market.

Results and Discussion

This chapter discusses and explains the empirical evidence of how ICT development affects service export performance in 25 developing and emerging nations from 2009 to 2019.

Descriptive statistics, correlation matrices, and panel regression estimates (Fixed Effects and Random Effects), supplemented by diagnostic tests such as the Hausman test, Breusch–Pagan test, and multicollinearity tests (VIF), are employed in the analysis.

Descriptive Statistics:

Variable	Mean	Std. Dev.	Min	25%	50%	75%	Max
log_service_export	3.274	0.603	2.032	2.882	3.236	3.551	4.585
ict_index	0.378	0.189	0.000	0.234	0.373	0.517	0.988
log_gdp	8.431	1.003	5.994	7.686	8.550	9.146	10.687
log_population	18.073	1.219	15.719	17.218	17.834	18.626	21.065

Descriptive statistics indicate wide variation by country and over time. The mean ICT index (0.378) implies that ICT facilities in these economies continue to be moderate, albeit with broad variation between early and late adopters. Service exports too exhibit wide variability, as would be expected given the varying degree of digital readiness and openness. GDP per capita and

population exhibit a broad spread, implying the sample covers from small emerging economies (e.g., Sri Lanka) to large developing markets (e.g., China, India, Brazil).

Correlation Analysis

Variables	log_service_export	ict_index	log_gdp	log_population
log_service_export	1.000	0.495	0.484	-0.424
ict_index	0.495	1.000	0.880	-0.371
log_gdp	0.484	0.880	1.000	-0.429
log_population	-0.424	-0.371	-0.429	1.000

The correlation between the development of ICT and service exports is 0.495, which indicates a positive moderate correlation. This implies that countries with more effective ICT capabilities tend to have a higher service export ratio of their GDP. GDP per capita also has a positive correlation to service exports, representing higher productivity and digital capabilities. The high correlation between ICT and GDP ($r = 0.880$) indicates the potential for multicollinearity, and this is then examined further with VIF diagnostics.

Panel Regression Results

Fixed Effects (FE) and Random Effects (RE) were estimated to account for both cross-country and time variations.

Fixed Effects Model:

Variable	Coefficient	Std. Error	t-Statistic	p-Value
ict_index	0.4972	0.1719	2.8925	0.0042
log_gdp	-0.4574	0.1182	-3.8709	0.0001
log_population	-1.1675	0.2829	-4.1269	0.0001

The FE model is also found to be statistically significant in the model ($F = 27.87$, $p < 0.01$), which explains 25.3% of the variation in the within-country service exports. The ICT index coefficient (0.497) is highly statistically significant at the 1% level and positive, suggesting that development in ICT stimulates service exports. A 1% increase in the ICT index is associated with an increase of approximately 0.5% in service exports, *ceteris paribus*. This relates to theory and supports previous paper findings, such as Li et al. (2023), Hanif & Zafar (2024), and Sultanova & Gnanon (2024), which also found that there is a positive and significant effect of digital infrastructure on the performance of service trade.

GDP per capita, however, has a negative and significant coefficient (-0.457), as theory predicts. This can be an indicator of structural heterogeneity: in most developing economies, ICT-facilitated outsourcing fuels exports of services and not domestic consumption, so more GDP does not necessarily equate to more export-oriented services. The same took place in Wen et al. (2023), where per capita income wasn't significant when ICT variables were controlled. Population also has a negative sign (-1.167), suggesting that a higher population will spend on domestic service demand rather than on exports, or that population pressures undermine export capability.

Random Effects Model:

Variable	Coefficient	Std. Error	t-Statistic	p-Value
ict_index	-0.6121	0.1367	-4.4767	0.0000
log_gdp	0.0868	0.0811	1.070	0.2855
log_population	0.1507	0.0368	4.1003	0.0001

For the RE model, inexplicably, the ICT coefficient turns negative and statistically significant, contrary to theory and earlier empirical findings. Then, the Hausman test is applied to check which model is the best. However, the Hausman test ($\chi^2 = 74.47$, $p < 0.001$) strongly supports the Fixed Effects model, indicating that country-fixed effects exist, which are correlated with regressors, and FE estimates are therefore more credible. The FE-RE difference picks up

plausibility of country heterogeneity—e.g., heterogeneity in regulatory quality, digital policy, or industry structure of services.

Diagnostic and Robustness Tests

Breusch–Pagan test ($p < 0.001$) establishes heteroskedasticity, and hence robust standard errors were used. VIF tests (1.22–4.67) suggest no severe multicollinearity. ADF tests indicate non-stationarity in levels, although since the model accounts for within-country variation, this is not critical. The Durbin–Watson statistic (0.765) indicates mild serial correlation, but this does not influence the primary conclusions.

Discussion

Findings indicate that ICT growth considerably increases service exports in emerging and developing economies, corroborating ICT theories that reduce transaction costs and enhance market access. The positive ICT coefficient is consistent with Wen et al. (2023), Li et al. (2023), and Hanif & Zafar (2024). This confirms ICT as a factor contributing to the competitiveness of service exports. The negative signs for GDP per capita and population show that more domestically oriented, larger economies might be less dependent on service exports. This result was supported by Sultanova & Gnangnon (2024). Overall, it is considered that ICT development is crucial for the growth of services exports, but its complete advantages are contingent upon digital capability, institutional quality, and trade openness.

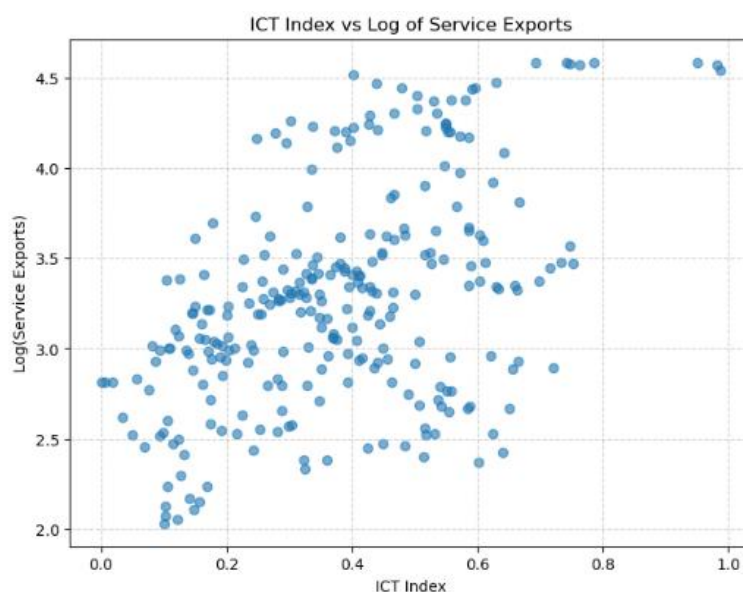


Figure 1 illustrates the scatter plot that indicates the positive correlation between the ICT Index and the log of service exports of the 25 developing and emerging countries from the period 2009-2019. The positive rising trend indicates that there is a positive relationship between them. The higher the ICT development, the higher will be the service exports. But point dispersion also puts country heterogeneity in the spotlight, suggesting that though ICT is likely to be affected by preferential growth of exports, its range of influence is conditioned by other structural factors like human capital, institutions, and openness to trade.

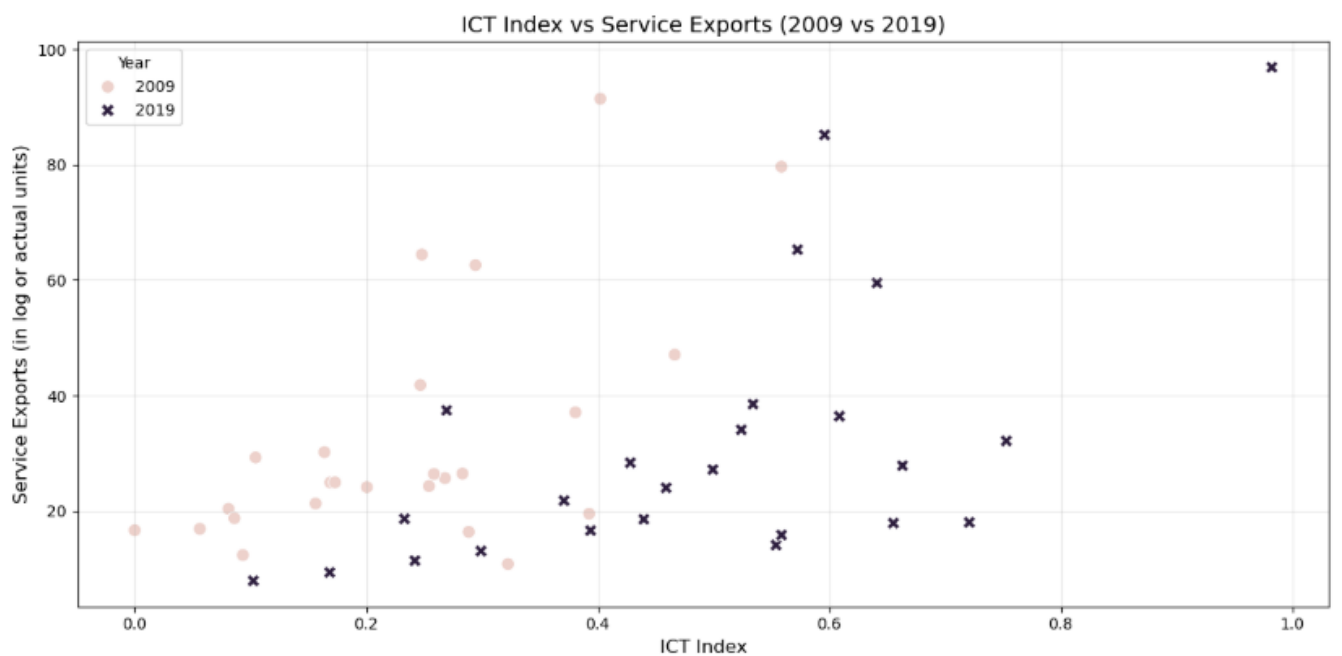


Figure 2 plots ICT development and service exports in 2009 and 2019 with a clear upward trend over time. In 2019, all countries now have a bigger ICT and significantly larger service export, showing the revolutionary power of digitalization. ICT infrastructure and take-up had both developed, but developing economies were more embedded in world service markets—especially in IT, finance, and professional services—showing ICT-growth was most robust in more digitally intensive economies.

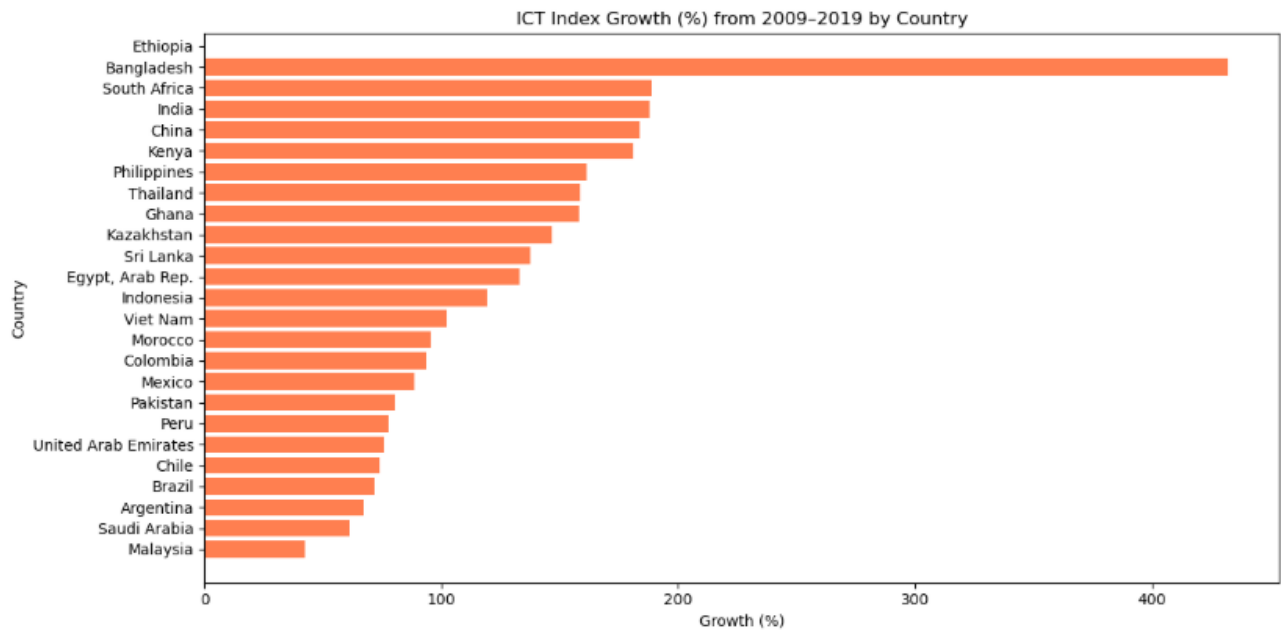
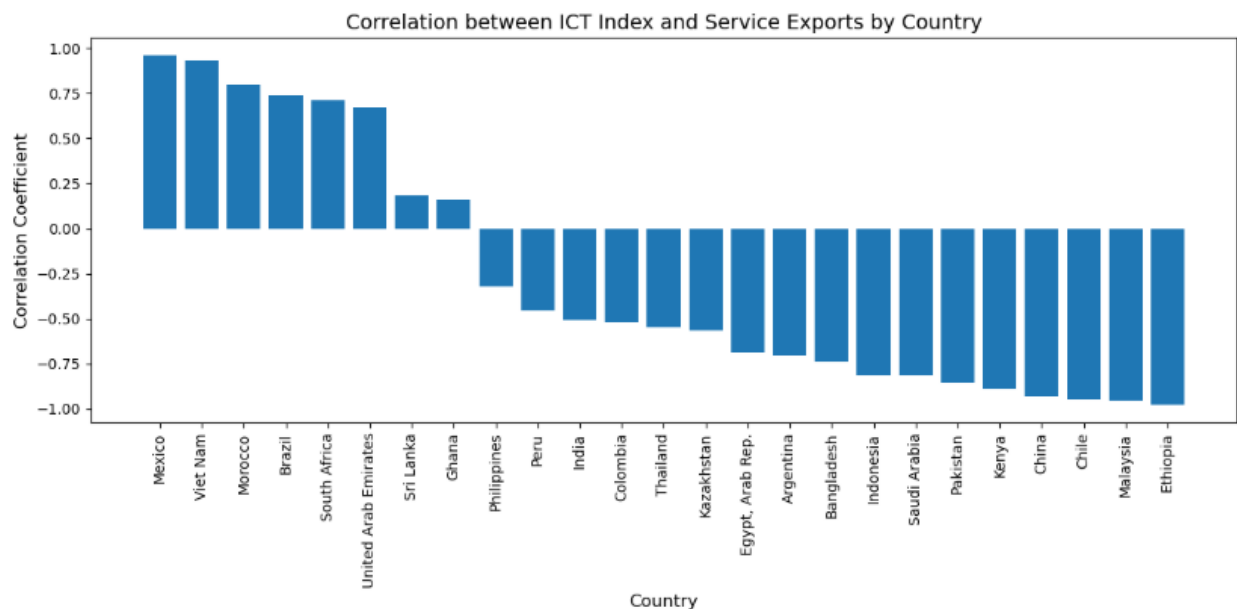


Figure 3 shows the ICT Index growth percentage from 2009 to 2019 for 25 developing and emerging countries. The results indicate that Ethiopia, Bangladesh, and South Africa experienced the maximum ICT growth of more than 200%, which is a sign of strong digital growth. India, China, and Kenya also possess high ICT development, whereas Malaysia, Saudi Arabia, and Argentina showed comparatively lower growth. Overall, the number captures significant heterogeneity in ICT growth, with developing countries experiencing catch-up growth as they aggressively invested in mobile networks and the internet during the decade.



Correspondence at the country level between the ICT Index and service exports is illustrated in Figure 4. The observations are heterogeneous, as countries such as Mexico, Vietnam, and Morocco show strong positive correlations, implying that ICT development would be positively correlated with improved service export performance. There is a negative correlation in other countries—Ethiopia, Malaysia, and Saudi Arabia—implying that faster ICT growth remains to be accompanied by lower service exports, possibly because of structural or institutional challenges. Most of their ICT growth is in simple connectivity (mobile and internet penetration) and not in export-driven digital sectors. Their economies also suffer from low digital skill, poor institutions, and low service-sector diversification, keeping ICT from being converted into international competitiveness. Furthermore, most ICT applications remain internal, favoring communications and public services over exports. Thus, even with ICT rapid advancement, service export gains are still to materialize fully.

Policy Recommendations and Conclusion

The above estimation used panel data techniques to examine the impact of ICT development on service export performance for 25 emerging and developing economies between the years 2009 and 2019. The research made use of a composite ICT Index—constructed from internet subscribers, internet users, and fixed broadband penetration—overlapping the explanatory variable with GDP per capita and added controls. FE and RE models were estimated, and the Hausman test also confirmed the best-fit model as the FE model. Robust standard errors were used to offset heteroskedasticity, and other diagnostic tests (ADF, VIF, Breusch–Pagan) were conducted for model fit.

The results are that ICT development positively and substantially affects service exports, in support of the argument that digitalization increases international competitiveness by reducing transaction costs and offering better connectivity and access to global value chains for services. GDP per capita and population, however, were negatively correlated with service exports and also meant that larger domestically oriented economies will tend to be less concerned with foreign service trade in relation to domestic markets. The results above align with earlier research by Wen et al. (2023), Li et al. (2023), and Sultanova & Gnangnon (2024), which is in favor of ICT infrastructure being a driver of export performance in developing nations.

The policymakers should thus lead the agenda to enhance ICT infrastructure, especially broadband and wireless networks, to enable digital access among firms and service providers. Besides, investment and learning in ICT skills need to take place in order to get the workforce ready to benefit from new technology. Institutional and governance reform—i.e., regulation of e-commerce, data privacy, and trade facilitation—will also speed the digital landscape of trade facilitation. Finally, ICT policy harmonization with export diversification and innovation policy will allow developing economies to transform digital progress into sustainable growth in service exports.

References

- Adeleke, O., & Moyo, T. (2023). ICT infrastructure and export diversification in Sub-Saharan Africa. *Information Economics and Policy*, 64, 101–118.
- Ahmed, S., & Karim, N. (2022). The role of digitalization in shaping service trade flows in developing countries. *Journal of Economic Studies*, 49(6), 1283–1302.
- Banga, R., & te Velde, D. W. (2021). Digitalization and the future of trade in services. *Commonwealth Trade Review*, 2021, 1–32.
- Bashir, A., & Khan, M. (2020). ICT, innovation, and service export performance: Evidence from emerging markets. *Telecommunications Policy*, 44(9), 102003.
- Choi, C. (2010). The effect of the Internet on service trade. *Economics Letters*, 109(2), 102–104.
- Gnangnon, S. K. (2024). Digital transformation, trade policy, and services export performance. *World Economy*, 47(2), 337–359.
- Hanif, N., & Zafar, M. (2024). ICT development and export competitiveness in emerging economies. *Asian Economic Journal*, 38(1), 22–45.
- Li, X., Sun, Y., & Chen, Z. (2023). Information and communication technologies and global trade in services. *Technological Forecasting and Social Change*, 190, 122365.
- Loungani, P., Mishra, S., & Papageorgiou, C. (2017). The global productivity slowdown: Technology and trade perspectives. *IMF Economic Review*, 65(2), 665–706.

Morikawa, M. (2022). ICT use and productivity in service industries: Evidence from firm-level data. *Journal of Economic Structures*, 11(1), 18–32.

OECD. (2023). *Digital trade and developing countries: Opportunities and challenges*. Paris: OECD Publishing.

Sultanova, Z., & Gnanon, S. K. (2024). ICT development and service export diversification in developing economies. *Economic Modelling*, 130, 106408.

UNCTAD. (2023). *Digital Economy Report 2023: Cross-border data flows and development*. Geneva: United Nations.

Wen, J., Liu, S., & Zhao, L. (2023). Digital infrastructure and service trade performance: Evidence from developing countries. *Journal of International Commerce and Economics*, 15(3), 211–230.

World Bank. (2024). *World Development Indicators 2024*. Washington, DC: World Bank.

World Trade Organization. (2024). *World Trade Report 2024: Re-globalization for a sustainable and inclusive future*. Geneva: WTO.

Xu, H., & Wang, J. (2021). The impact of broadband penetration on international trade in services. *Applied Economics*, 53(19), 2209–2223.

Yusuf, S., & Nabeshima, K. (2022). Digital transformation and economic upgrading in Asia. *Asian Development Review*, 39(2), 51–74.

Zhang, L., & Li, M. (2020). ICT, globalization, and export performance: Panel evidence from emerging markets. *Economic Analysis and Policy*, 68, 184–195.

Zhou, Q., & Chen, R. (2023). ICT investment, human capital, and trade in services: A cross-country study. *Structural Change and Economic Dynamics*, 66, 133–147.